MLP for Tabular Medical Data

QLS–MiCM Workshop

Why Compare MLPs to Linear Models?

Medical tabular data often has:

- few features.
- moderate sample size.
- mostly linear or monotonic relationships.

Logistic regression is often hard to beat.
MLPs help **only if** the data contain meaningful **nonlinear and/or interaction patterns**.

This exercise tests that idea.

What an MLP Can Learn

A multilayer perceptron can model:

- nonlinear boundaries.
- interactions between features.
- complex patterns missed by linear models.

But can also:

- overfit easily.
- require tuning (activation, layers, regularization).
- fail when data is simple or small.

You will explore both outcomes.

The Baseline: Logistic Regression

Logistic regression is:

- linear.
- stable.
- interpretable.
- often the best model.

What You Will Change in the MLP

You will experiment with:

🔧 Architecture

- number of layers `(e.g., (8,), (32,16), (100,100))`
- number of neurons.

🔧 Activation Functions

- `relu`.
- `tanh`.
- avoid `identity` for hidden layers

🔧 Regularization

- adjust `alpha`.
- understand it acts like L2 weight decay.

🔧 Training Settings

- increase `max_iter`.

How You Will Evaluate

Using `classification_report`:

- **Accuracy** — correctness.
- **Precision** — false-alarm.
- **Recall** — disease-detection.
- **F1-score** — balance of precision + recall.

Key question:.

Does added model complexity improve clinically meaningful metrics?